

Revisiting Rumelhart and McClelland (1986) for Romanian Verb Conjugation

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Abstract

This project reproduces the classic experiment of Rumelhart and McClelland (1986) on learning past tense verbs, adapted for the Romanian language. Unlike the original study, we employ a modern **Stacked Bidirectional LSTM** architecture with character-level embeddings. We significantly extended the scope of the experiment by training on a dataset of approximately **15,000 examples**, covering both Participle and Imperfect tenses. We demonstrate that the model can generalize to unseen verbs (including invented ones) with **97.57% accuracy**, effectively solving the "dual-route" mechanism debate using a single neural network.

1 Introduction

The core problem addressed in this project is the acquisition of grammatical rules by neural networks. In 1986, Rumelhart and McClelland (1) demonstrated that a neural network could learn English past tense rules (adding "-ed") without explicit programming. We aim to investigate if this holds true for Romanian, a Romance language with complex conjugation patterns involving suffixes (e.g., *-at*, *-it*) and significant irregularities.

Our main contributions are:

- Adaptation of the 1986 experiment for Romanian morphology.
- Implementation of a robust **Synthetic Data Generator** to overcome data scarcity.
- Comparison of generalization capabilities using a modern **Stacked Bidirectional LSTM** architecture.

We chose this project to understand the transition from rote memorization to rule generalization in deep learning models.

2 Approach

We implemented a character-level Sequence-to-Sequence model using Python, Keras, and TensorFlow. The code and dataset are available in our repository: https://github.com/Victorle/Project_Rumelhart_Romanian

2.1 Data Generation

Since a large corpus of Romanian verb pairs is not readily available, we created a **Synthetic Data Generator**. This algorithm creates pseudo-random verb roots adhering to Romanian phonotactics (C-V-C structure). Crucially, for each verb, we generated the full conjugation paradigm: **Participle plus 6 forms of Imperfect**, resulting in a total dataset of approximately **15,000 examples**. We combined these synthetic verbs with real irregular verbs to create a balanced training corpus.

2.2 Model Architecture

Instead of a simple MLP or a single LSTM, we used a **Stacked Recurrent Neural Network** (Stacked RNN) specifically designed for deep sequence processing:

1. **Embedding Layer:** Maps input characters to a **128-dimensional** dense vector space (increased from standard 64).
2. **Stacked Bidirectional LSTM:** We used a deep architecture with two stacked Bi-LSTM layers (**256 and 128 units respectively**). This allows the model to capture both raw phonological features and abstract grammatical rules.
3. **Output:** A TimeDistributed Dense layer with Softmax activation predicts the character probability distribution.

Training took approximately **30 minutes** on Google Colab GPU (NVIDIA T4), converging efficiently on the massive dataset of 15,000 examples.

3 Experimental Results

The model achieved a spectacular **97.57% accuracy** on a strictly held-out test set of 200 "Alien" verbs (1,400 total forms).

Key Findings:

- **Complexity Mastery:** The model achieved **100% accuracy** on complex phonological rules (e.g., roots ending in *-ge*, *-ce*, *-ne* transforming to *-s*), refuting Pinker's critique regarding internal structure changes.
- **Paradigm Stability:** Surprisingly, accuracy on the Imperfect tense (**98.00%**) was higher than the Participle (**95.00%**).
- **Error Analysis:** Errors were mostly "over-regularizations" (e.g., predicting *hikuris* instead of *hikuriut*), mimicking the "U-shaped learning" curve.

4 Limitations

The primary limitation is the use of synthetic data. While effective for learning abstract rules, the generated verbs do not reflect the semantic distribution of real-world language. Additionally, the model operates strictly at the character level and does not account for sentence-level context.

5 Conclusions

We successfully reproduced and extended the connectionist experiment for Romanian. The **Stacked Bidirectional LSTM** proved superior to traditional networks, demonstrating that a single neural mechanism can learn both regular rules and complex exceptions without explicit programming.

References

- [1] David E. Rumelhart and James L. McClelland. 1986. *On learning the past tenses of English verbs*. MIT Press.